Lumis

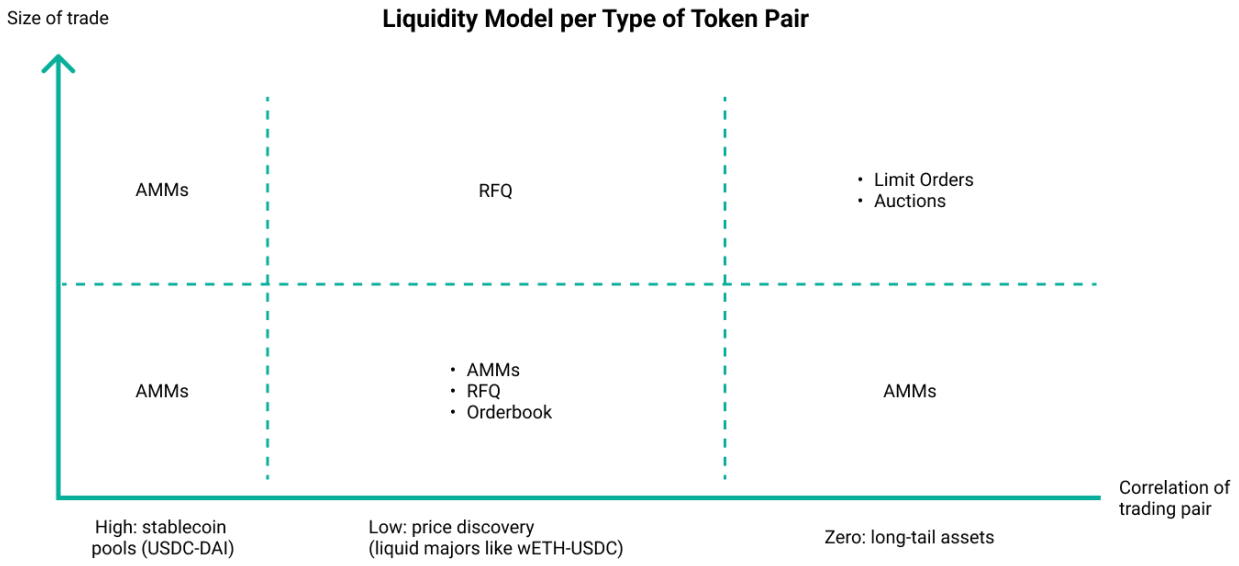
Atomic derivatives protocol

Introduction

This paper introduces atomic derivatives - a new type of financial primitives that by leveraging intent-centric architecture enables targeting any arbitrary portfolio profiles and risk preferences. It can be vaults that recursively lend/borrow assets to get leveraged token exposure without liquidations, DCA, convex option-like payoff, basis trade, or even interlinked vaults that match directional risk-takers and risk-averse persons by respectively replicating call-option and put-option-like payoff to provide liquidity-as-a-service product that bootstraps liquidity for both spot and derivatives markets while ensuring targeted strategy profiles aiming to outperform either base or quote asset.

Background

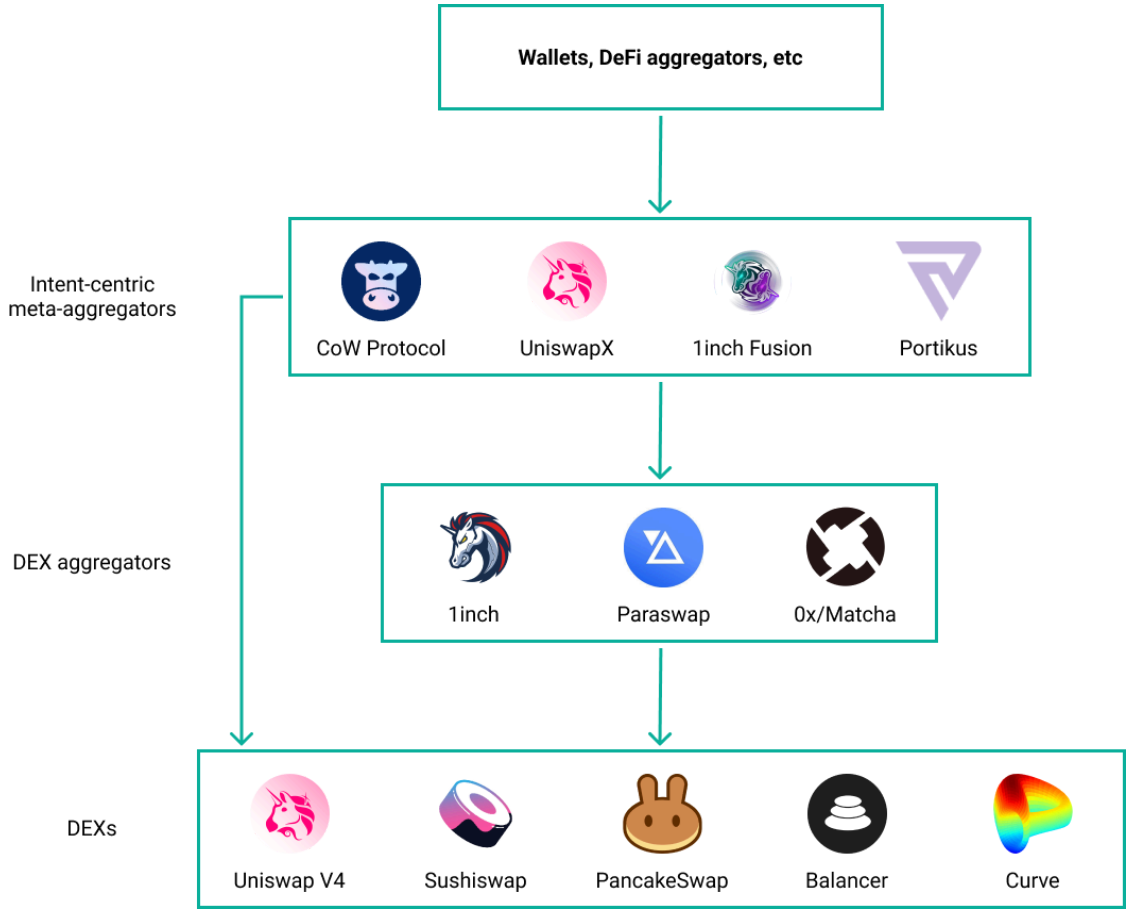
Since the introduction of the "xyk" model pioneered by Uniswap V1/V2, the on-chain liquidity market structure has significantly evolved. While groundbreaking, the monolithic approach of constant product AMM had limitations, such as high slippage and inefficient liquidity usage. As the space matured, new DEX architectures emerged, tailored to specific use cases, such as the Stableswap invariant from Curve Finance, the Request-for-Quote (RFQ) system from 0x, batch auctions from CoWSwap, and the concentrated liquidity model of Uniswap V3. Each closes its own unique market needs by improving liquidity providers' user experience in terms of flexibility and/or capital efficiency.



These advancements mirror the evolution of blockchain technology itself. As Ethereum was the first who separate consensus and application logic, enabling developers to create dApps

without building their own blockchains but instead leveraging a shared security model, the latest DEX models began to separate liquidity provision from clearing and settlement. In this regard, Uniswap V3 custom liquidity ranges provide LPs greater flexibility in terms of expressing their portfolio preferences and boosting capital efficiency while still receiving shared trade flow from aggregated liquidity curves.

Similarly, as Ethereum moved to a modular scalability approach by separating execution and settlement/validation via applying rollups + data availability layer model - we are witnessing further trends of externalizing management from financial infrastructure in both DEXes and lending markets architectural landscapes. Namely the Coincidence-of-Wants (CoWS) mechanism from CoW Swap and the emergence of intent-centric architectures like CoW Protocol intents/hooks and Uniswap V4 hooks in decentralized exchanges DeFi segment.



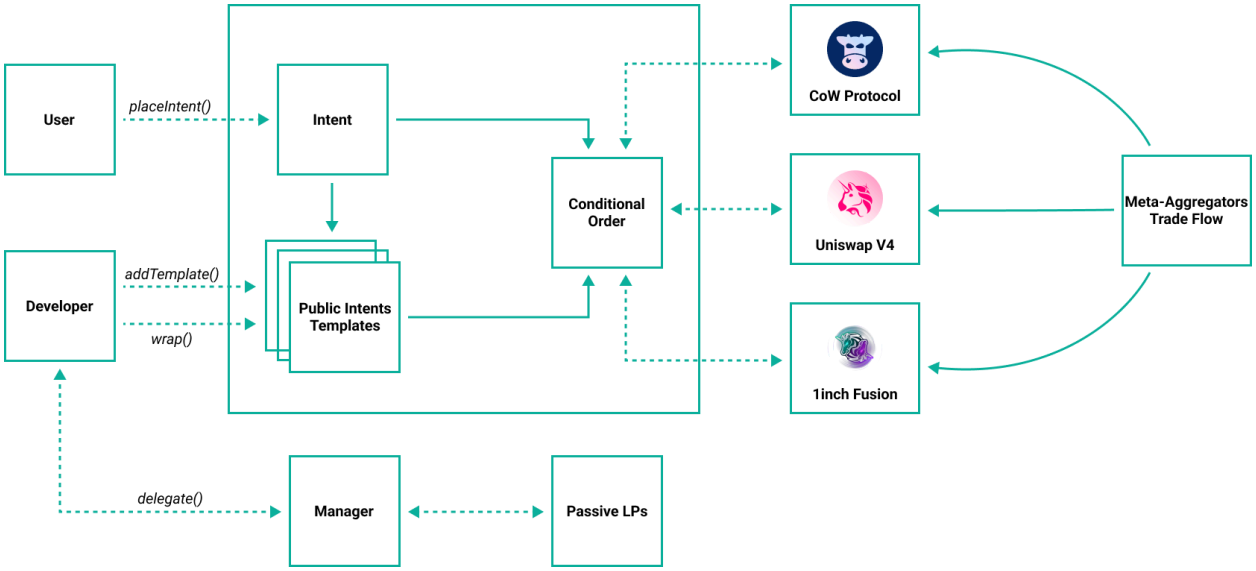
This approach enables LPs to express their preferences and requirements through intents, which can include parameters like price, size, and customized trading strategies via conditional orders. The routing problem is being outsourced to external parties - a network of solvers or fillers who compete to satisfy these intents by sourcing liquidity for optimal order execution. This separation allows LPs to focus on defining their desired trading strategies and risk profiles without restrictions to follow any specific rules such as Uniswap V3 invariant curve - enabling

ultimate flexibility in adapting strategies algorithms to constantly evolving market conditions. At the same time, inclusion into a competitive solvers network ensures preserving trade flow by tapping into the liquidity and user base of established meta-aggregators.

Protocol

Targeting optimal trading execution and strategies inclusion into competitive solvers networks, we introduce conditional liquidity protocol that enables expressing intents compatible with all meta-aggregators, such as CoW Protocol, Uniswap X, Paraswap, and 1inch Fusion.

Inspired by CoW AMM design, it employs a programmatic order framework by CoW Protocol to deploy conditional orders that express liquidity providers' intents and desires to achieve specific portfolio requirements.



1) Intent specification: the user selects one of the available intent templates or specifies their own intent requirements
2) Order deployment: once intent is expressed, we deploy it as a conditional order following the programmatic order framework by CoW Protocol
3) Intent translation: protocol automatically translates it to the app-specific compatible formats, such as Uniswap V4 hooks (NoOp modification), Paraswap, etc.
4) Vault: once intent is deployed, the intent owner can wrap it into a vault and delegate its ownership to a manager contract - enabling this way liquidity deposits from external LPs.

Such architecture with choosing CoW Protocol for the primary deployment instead of Uniswap V4 hooks allows us to get unique features of programmatic order frameworks, specifically Wait4CoW orders that only match an order with other CoW traders. This way, by compatibility with both Uniswap V4 hooks and CoW Protocol programmatic orders we can specify, for

example, accepting trade flow only from Uniswap X fillers that come from Uniswap front-end while ensuring rebalances execution only as a part of CoW Protocol batch auction - targeting this way slippage and overall market impact minimization.

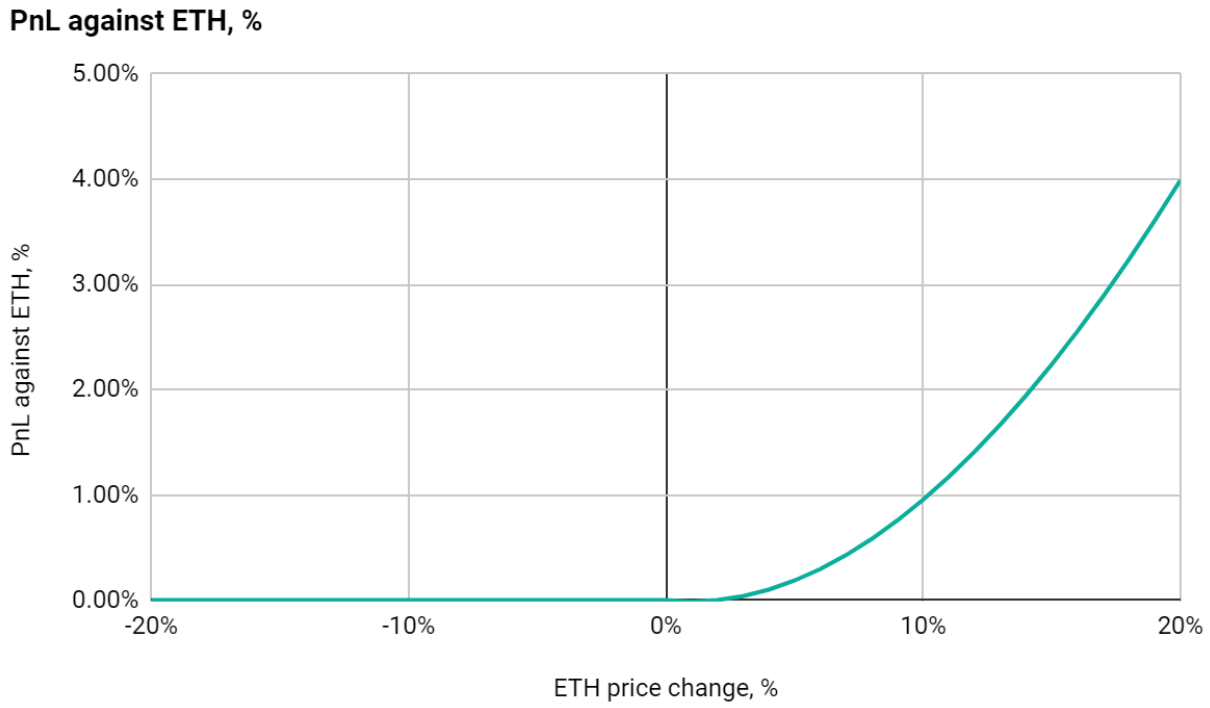
Atomic derivatives

By leveraging this intent-centric architecture we can create vaults that target any arbitrary portfolio payoff and risk structure. Resulting from the complexity of precise Uniswap V3 LP-positions hedging, atomic derivatives apply dynamic position adjustments by gradual positions modifying responding to evolving market conditions in an atomic way - internalizing this way potential MEV opportunities from various market discrepancies.

For example, let's consider a vault that provides liquidity following the Uniswap V3 curve and dynamically hedges it with long power perpetuals.

The vault accepts ETH and splits it into 2 parts:

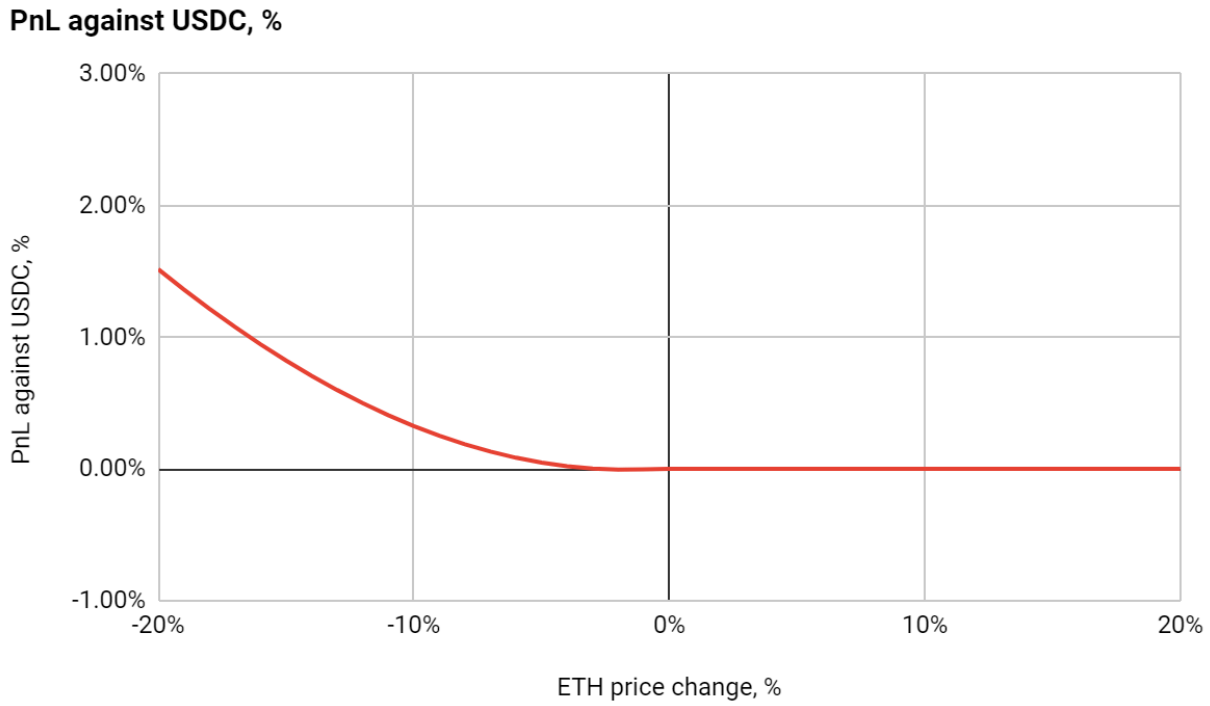
1) 50% to provide liquidity for the ETH-USDC pool on the upside only (1.0x - 1.2x)
2) 50% posted as collateral on lending protocols to gradually borrow more and more USDC to increase long power perpetuals position (long ETH^2 or oSQTH) as ETH/USDC continues to grow.



This combination results in a call-option-like payoff against ETH enabling ETH outperformance even during the craziest ETH price movements. As we provide liquidity on the upside only - the Uniswap AMM mechanism on the downside moves fully converting the position into ETH while

our USDC debt is zero. At the same time, as the ETH price grows more and more from the initial point - we acquire more USDC debt that is equal to the USDC amount received by the AMM from the ETH selling and use this USDC debt to buy more long oSQTH - precise atomic hedging via on-demand DCA.

Similarly, by mirroring the first vault - we can create a vault that offers a put-option-like payoff against USDC.



This vault accepts USDC and split it into 2 parts:

1) 50% to provide liquidity for the ETH-USDC pool on the downside only (0.8x - 1.0x)
2) 50% to post it as collateral to gradually borrow more ETH as the price continues to go down and use this ETH to increase our short ETH power perps position

As both of the vaults are capped at a maximum LTV of 90% after which we can't borrow more funds to acquire more hedge to preserve the desired convexity of our payoff - we execute conditional rebalances during which we adjust portfolio proportion to bring back the initial portfolio composition. Including our vault rebalances to batch trades from CoW Protocol optimizes rebalance execution enabling the smooth rolling of our positions to the next trading epoch in a similar manner to how perpetual futures work.

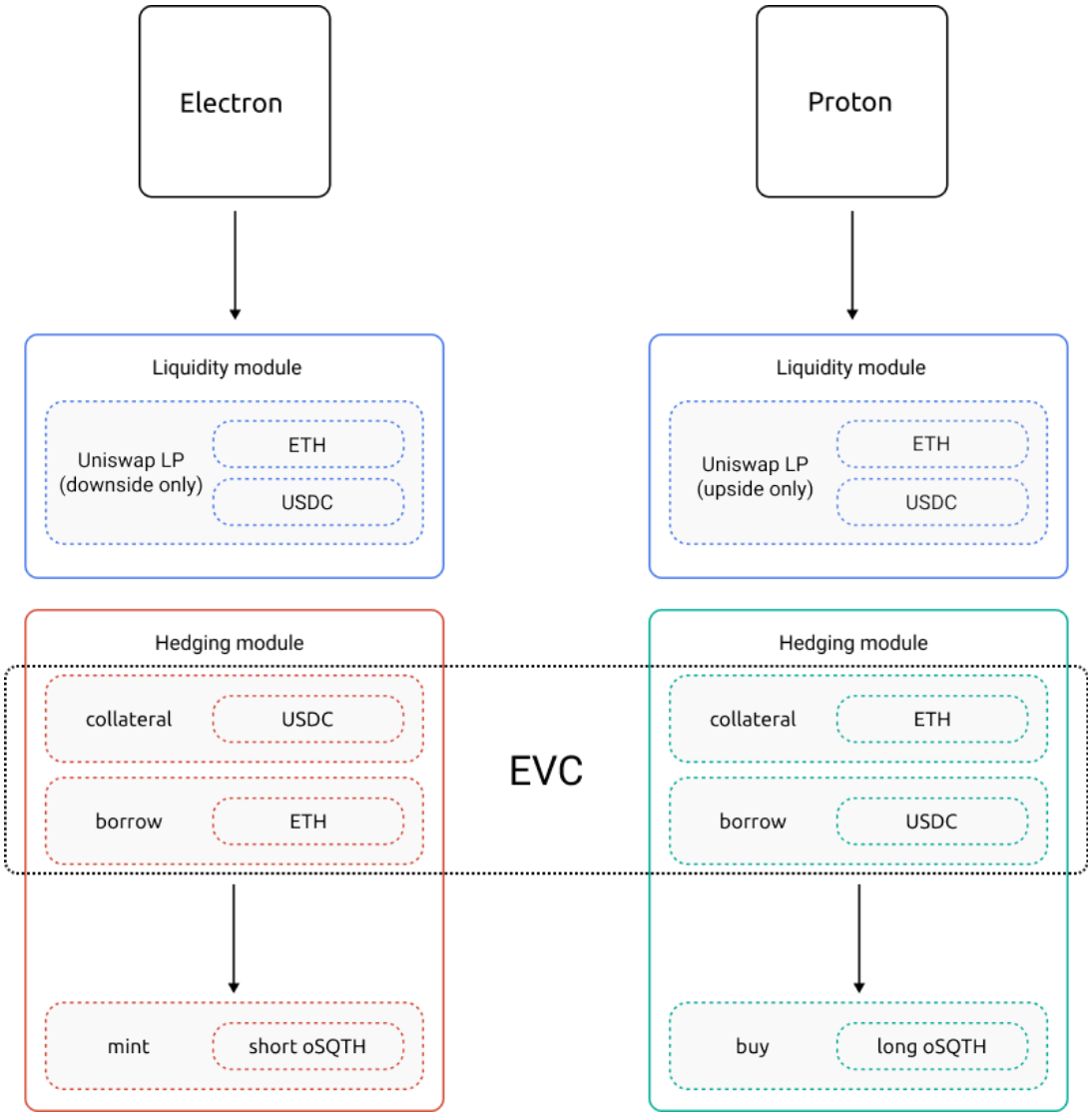
Operating on L2s with private mempools where adversarial actors cannot sandwich our subject-for-frontrunning hedging operations with power perps position - with atomic hedging our mechanism internalizes MEV arises between ETH and theoretical oSQTH price discrepancies.

Similarly, we can create and facilitate vaults with any arbitrary payoffs:

1) Recursive perps - recursively borrow/lend assets to get leveraged exposure;
2) AMMs with custom logic, such as CSMM, LAAMBert, etc;
3) Custom price curves - x^2, x^n, sin(x), etc;

Liquidity-as-a-service

One of the examples of vaults with arbitrary logic is a mechanism that matches directional risk-takers and risk-averse persons by respectively replicating call-option and put-option-like payoffs described earlier to create a liquidity-as-a-service system that can be used for bootstrapping on-chain liquidity.



The mechanism consists of Proton and Electron vaults that follow previously described call and

put atomic options templates respectively. Vaults are interlinked in a closed system through Ethereum Vault Connector (EVC) - an isolated lending/borrowing market in which only Proton and Electron are allowed to lend/borrow funds from each other.

The system targets equilibrium through the Jump-Rate Curve mechanic. This way, similarly to the perpetual futures funding rate mechanism - during the significant price deviation from the initial points - the borrowing rate will rapidly increase, incentivizing the opposite side to provide more liquidity as the yield can be accessible only by depositing into the opposite vault.

On the downside Proton (call) fully consists of ETH - the opposite Electron (put) borrows this ETH to mint more short oSQTH effectively paying for it the borrowing rate. While the Electron (put) vault on the upside fully consists of USDC lending it to the opposite Proton (call) vault so it can buy more long ETH power perpetuals while paying for it the borrowing rate determined by traditional Jump-Rate curve model optimized for our purposes.

This way, we separate upside and downside liquidity management between persons based on their portfolio preferences. With this approach, we enable investors to get precise hedging of their LP position while preserving trading fees - allowing them to outperform either base or quote assets. Investors who are looking for specific asset outperformance, for example, ETH, deposit to the Proton vault receiving call-option-like payoff against ETH. Investors who prefer delta-neutral exposure can select Electron vault receiving a put-option-like payoff.

With permissionless power perpetual market listing on the upcoming PerpWorld product from Opyn, the proposed mechanism can be deployed for any ERC-20 token offering a liquidity-as-a-service product that bootstraps both spot and derivatives markets.

Tailored by specific token market structures and project needs - the mechanism can be modified to use alternative liquidity curves and hedging methods. Intent-centric architecture with inclusion into competitive solvers networks allows us to get rid of invariant-based liquidity curves such as Uniswap V3 - enabling greater flexibility and capital efficiency.

Additionally, each vault can be expanded by integrating external modules, such as:

1) Zk-coprocessors: Axiom, Lagrange, etc;
2) Gas abstraction: Gazelle, UMA etc;

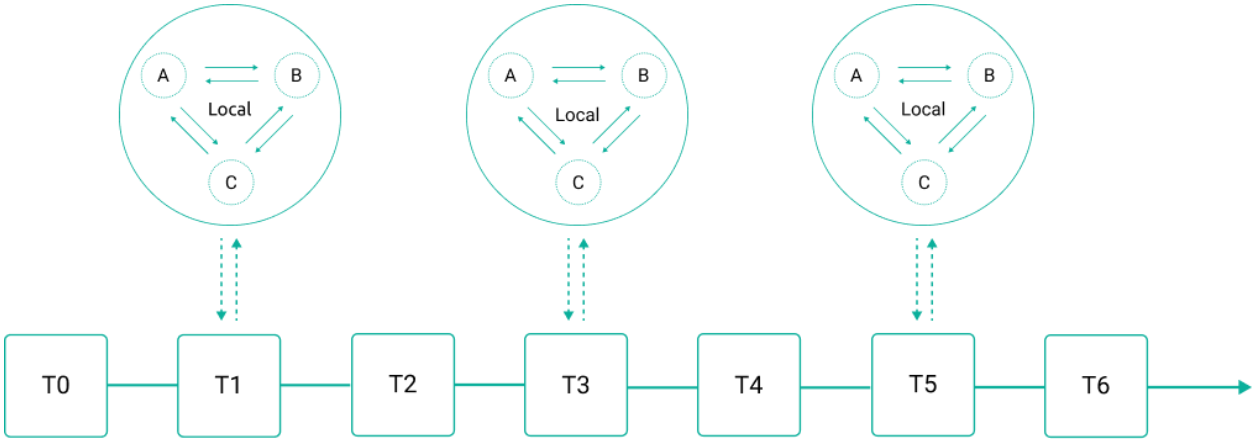
Improving this way the decision-making process by leveraging off-chain computations and optimizing gas expenses.

Vaults network

The intent-based protocol architecture enables permissionless vault rebalance coordination. By introducing global trading epochs - each vault can target conditional time rebalance inclusion

into specific batches based on different time frames, such as hourly, daily, weekly, and monthly, depending on the specific requirements of the vault strategy.

Resulting from the complexity of manual rebalance synchronization between multiple individual vaults - this approach provides developers with a flexible schedule of discrete auctions aiming for value surplus maximization.



This way, developers can, for example, match recursive leverage strategies during exposure adjustments not only locally against each other but also include them into global trading epochs either public or private (permissioned) to improve trading strategies execution and risk management. This leads to a liquidity network effect that benefits both spot traders and atomic vault developers.

Conclusion

The future of liquidity is modular. By leveraging an intent-centric architecture, the Lumis protocol enables the creation of atomic derivatives - vaults with arbitrary logic that can express and execute complex, user-defined strategies. This approach opens up a wide design space for novel DeFi primitives and on-chain liquidity management models, from non-liquidatable leveraged positions to custom AMM curves. Atomic derivatives offer ultimate flexibility and capital efficiency, while compatibility with all meta-aggregators ensures inclusion into competitive solver networks for preserved trade flow.